Customer Churn Prediction

Data Gathering, Exploratory Data Analysis & Preprocessing Report

Prepared for: Data Science Project

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Dataset: Customer_Churn_Data_Large.xlsx

1. Data Gathering & Dataset Selection

This project uses a single Excel workbook containing five interconnected sheets. Each sheet provides a distinct dimension of customer behaviour. All five were selected because together they cover the three key domains known to drive churn: who the customer is (demographics), what they do (transactions and online activity), and how they interact with support (customer service).

1.1 Dataset Overview

Sheet Name	Rows	Key Columns	Role
Customer_Demographics	1,000	Age, Gender, MaritalStatus, IncomeLevel	Baseline customer profile
Transaction_History	5,054	AmountSpent, ProductCategory, TransactionDate	Purchase behaviour
Customer_Service	1,002	InteractionType, ResolutionStatus	Support interactions
Online_Activity	1,000	LoginFrequency, LastLoginDate, ServiceUsage	Digital engagement
Churn_Status	1,000	ChurnStatus (0/1)	Target variable

1.2 Rationale for Inclusion

- Customer_Demographics: Provides baseline segmentation variables. Age, income level, and marital status are well-established predictors of service switching behaviour.
- Transaction_History: Recency, frequency, and monetary value of purchases are the foundation of RFM analysis — one of the strongest frameworks for predicting churn.
- Customer_Service: Complaint frequency and unresolved issues are direct indicators of customer dissatisfaction, a primary churn driver.
- Online_Activity: Login frequency and days since last login measure engagement. Low engagement is a leading indicator of disengagement before formal churn.
- Churn_Status: The target variable. Binary label (1 = churned, 0 = retained) required for supervised classification model training.

1.3 Merging Strategy

All five sheets share CustomerID as the primary key. Transaction_History and Customer_Service contain multiple rows per customer and must be aggregated before merging. All merges use LEFT JOIN with Customer_Demographics as the anchor table to ensure all 1,000 customers are retained regardless of whether they appear in supplementary sheets.

Sheet	Before Merge	After Merge	Notes
Transaction_History	5,054 rows	1,000 rows	Aggregated: sum, mean, count, mode, max date
Customer_Service	1,002 rows	1,000 rows	332 customers had no record — NaN filled with 0
Online_Activity	1,000 rows	1,000 rows	Direct merge — no aggregation needed
Churn_Status	1,000 rows	1,000 rows	Direct merge — target variable

2. Exploratory Data Analysis

2.1 Descriptive Statistics

Following the merge, the master dataset contains 1,000 rows and 17 columns with zero missing values. The table below summarises key statistics for all numeric features.

Feature	Mean	Median	Std Dev	Min	Max	Skew
Age	43.3	43.0	15.2	18	69	Low
TotalSpend (£)	1,197	1,136	668	9.8	3,386	Low
AvgSpend (£)	248.8	250.1	79.4	9.8	497.0	Low
NumTransactions	5.05	5.0	2.6	1	9	Low
LoginFrequency	25.9	27.0	14.1	1	49	Low
DaysSinceLastLogin	181.4	181.5	105.7	1	365	Uniform
NumComplaints	0.33	0.0	0.49	0	2	Right

Categorical Feature Summary

Column	Distribution
Gender	Female: 513 (51.3%) | Male: 487 (48.7%)
MaritalStatus	Widowed: 276 (27.6%) | Married: 261 (26.1%) | Divorced: 248 (24.8%) | Single: 215 (21.5%)
IncomeLevel	High: 349 (34.9%) | Medium: 326 (32.6%) | Low: 325 (32.5%)
ServiceUsage	Online Banking: 349 (34.9%) | Mobile App: 342 (34.2%) | Website: 309 (30.9%)
FavCategory	Books: 313 (31.3%) | Clothing: 226 (22.6%) | Electronics: 182 (18.2%)
ChurnStatus	Retained (0): 796 (79.6%) | Churned (1): 204 (20.4%)

2.2 Visualisations

Figure 1 shows the distribution of key numeric features and churn rates across demographic segments.

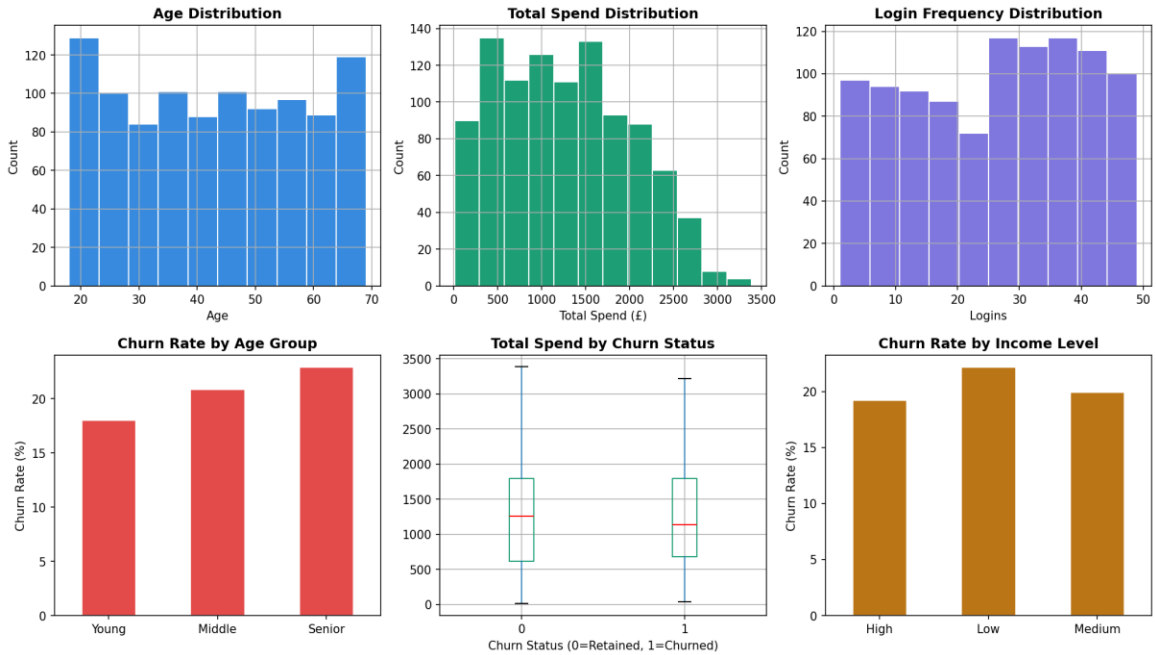

Figure 1: Feature distributions and churn rates by demographic group

Figure 2 shows the churn distribution and breakdown by gender and marital status.

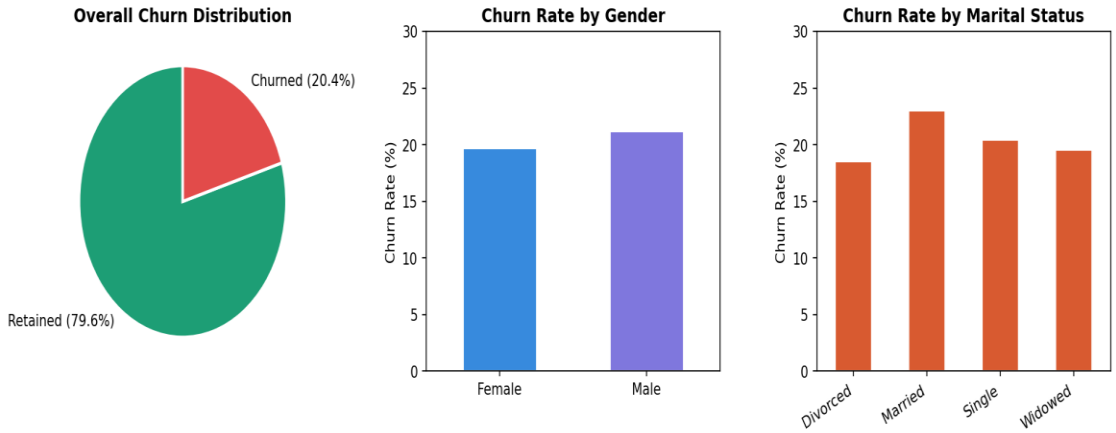

Figure 2: Churn distribution and breakdown by key demographic variables

2.3 Correlation Analysis

After encoding categorical variables, Pearson correlation coefficients were calculated for all numeric features against ChurnStatus. Figure 3 shows the full correlation matrix.

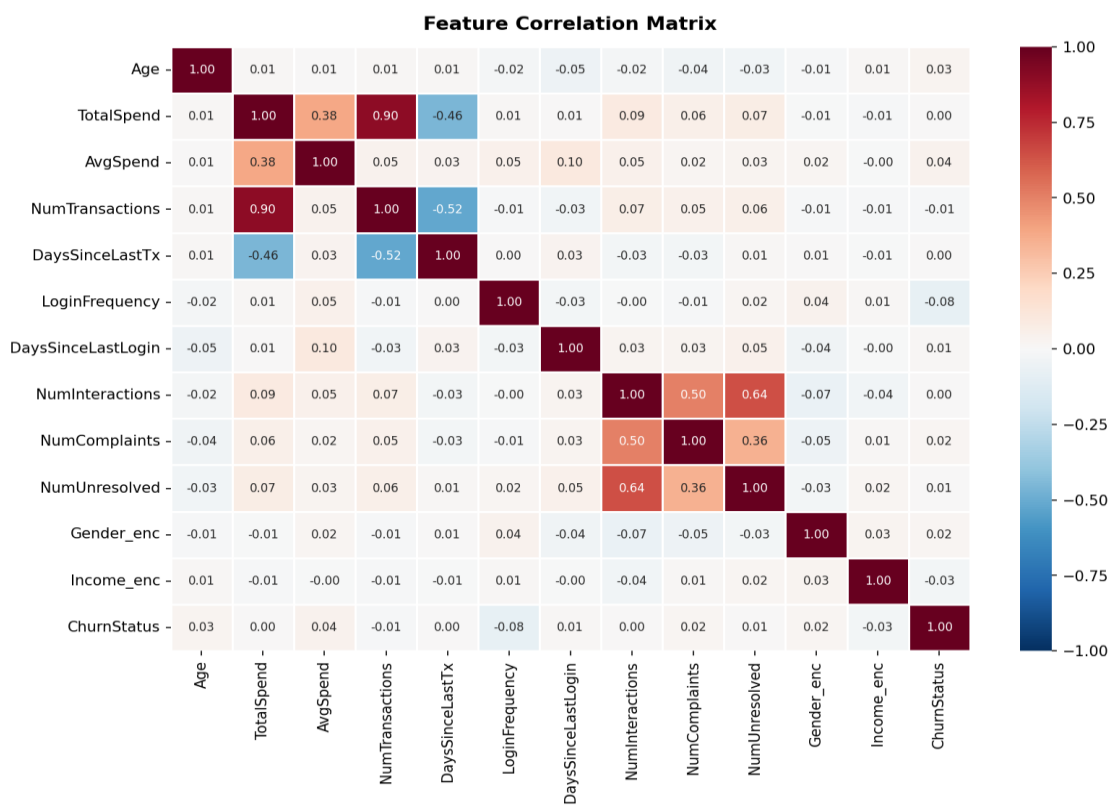

Figure 3: Feature correlation matrix

Key Correlation Findings

Feature	r vs Churn	Interpretation
LoginFrequency	-0.082	Strongest predictor — low engagement precedes churn
AvgSpend	+0.045	Higher average spend slightly associated with churn
Income_enc	-0.030	Lower income customers churn marginally more
Age	+0.029	Older customers churn slightly more than younger
NumComplaints	+0.017	More complaints weakly associated with higher churn

Note: All individual correlations are weak (|r| < 0.1). This is typical in real-world churn datasets where churn is driven by combinations of features rather than any single variable. Tree-based ensemble models such as Random Forest and XGBoost, which capture feature interactions, are expected to outperform linear classifiers on this dataset.

2.4 Key Insights from EDA

- Churn rate is 20.4% — a moderately imbalanced binary target. Standard accuracy is not a reliable evaluation metric; AUC-ROC and F1-score must be used.
- Senior customers (56-69) churn at 22.9% versus 18.0% for young customers (18-35), suggesting age is worth retaining as a feature.
- Low-income customers churn at 22.2% versus 19.2% for high-income, a modest but consistent pattern.
- Login frequency is the single strongest individual predictor (r = -0.082). Customers with fewer logins are more likely to churn.
- 332 customers (33.2%) had no customer service record — their absence of contact may itself be an informative signal of disengagement.
- Product category preferences are relatively uniform, suggesting category alone is not a strong churn differentiator.

3. Data Cleaning & Preprocessing

3.1 Missing Value Treatment

The raw data contained zero missing values across all five sheets. However, the LEFT JOIN merge introduced 332 NaN values in the Customer_Service derived columns for customers who had no service interaction record. These were handled using the flagging and fill strategy described below.

Column	NaN Count	Strategy	Justification
NumInteractions	332	Fill with 0	No contact record = 0 interactions
NumComplaints	332	Fill with 0	No contact record = 0 complaints
NumUnresolved	332	Fill with 0	No contact record = 0 unresolved
HasServiceRecord	N/A	New binary flag	1 = had contact, 0 = never contacted support

3.2 Outlier Detection & Treatment

The IQR method (Tukey fences: Q1 - 1.5*IQR and Q3 + 1.5*IQR) was applied to all numeric features. Figure 4 shows box plots for three key numeric columns.

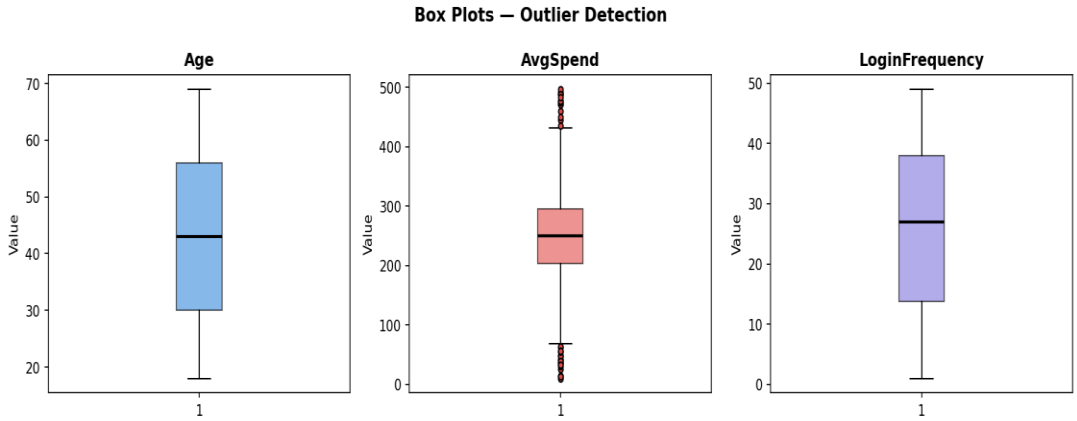

Figure 4: Box plots for Age, AvgSpend, and LoginFrequency

Feature	Low Outliers	High Outliers	Decision	Rationale
Age	0	0	No action	Range 18-69 is naturally bounded
TotalSpend	0	0	No action	Range reflects legitimate variation
AvgSpend	23	15	Cap (Winsorise)	Driven by single-transaction customers; clip to 5th-95th percentile
NumTransactions	0	0	No action	Range 1-9 is expected
LoginFrequency	0	0	No action	Range 1-49 is expected
DaysSinceLastLogin	0	0	No action	Range 1-365 is naturally bounded

3.3 Categorical Encoding

All categorical columns were converted to numeric form using the encoding method most appropriate for each column's data type.

Column	Method	Output	Justification
Gender	Label encoding	M=1, F=0	Binary — no ordering issue
MaritalStatus	One-hot (drop_first)	3 binary columns	4 nominal categories — no natural order
IncomeLevel	Ordinal encoding	Low=0, Med=1, High=2	Natural order exists
ServiceUsage	One-hot (drop_first)	2 binary columns	3 nominal categories — no natural order
FavCategory	One-hot (drop_first)	4 binary columns	5 nominal categories — no natural order

3.4 Feature Engineering

Five new features were created to capture relationships not directly expressed by the raw columns.

Feature	Formula	Purpose
AgeGroup_enc	pd.cut(Age, bins=[17,35,55,70])	Captures non-linear relationship between age and churn
SpendPerTx	TotalSpend / NumTransactions	Spend intensity per visit, independent of transaction volume
ComplaintRate	NumComplaints / NumInteractions	Proportion of contacts that were complaints — more nuanced than raw count
IsHighRisk	LoginFrequency<10 AND NumComplaints>0	Combines two weak signals into a single high-risk flag (51 customers)
HasServiceRecord	NumInteractions > 0	Preserves the informational value of never having contacted support

3.5 Normalisation

All 12 numeric feature columns were standardised using Z-score standardisation (StandardScaler) to give each feature a mean of 0 and standard deviation of 1. This ensures no single feature dominates model training due to scale differences. Binary and ordinal encoded columns were not scaled as they are already on a bounded scale.

4. Final Cleaned Dataset

4.1 Dataset Summary

Total rows	1,000
Total features (excl. target)	26
Target variable	ChurnStatus (0 = retained, 1 = churned)
Missing values	0
Numeric features (scaled)	12
Engineered features	5
Encoded categorical features	9
Churn rate	20.4% (204 churned / 796 retained)

4.2 Feature List

Feature	Type	Source / Notes

Age	Numeric (scaled)	Customer_Demographics
TotalSpend	Numeric (scaled)	Aggregated from Transaction_History
AvgSpend	Numeric (scaled)	Aggregated — Winsorised at 5th/95th percentile
NumTransactions	Numeric (scaled)	Aggregated from Transaction_History
DaysSinceLastTx	Numeric (scaled)	Engineered from TransactionDate
LoginFrequency	Numeric (scaled)	Online_Activity
DaysSinceLastLogin	Numeric (scaled)	Engineered from LastLoginDate
NumInteractions	Numeric (scaled)	Aggregated from Customer_Service
NumComplaints	Numeric (scaled)	Aggregated — NaN filled with 0
NumUnresolved	Numeric (scaled)	Aggregated — NaN filled with 0
SpendPerTx	Numeric (scaled)	Engineered: TotalSpend / NumTransactions
ComplaintRate	Numeric (scaled)	Engineered: NumComplaints / NumInteractions
Gender_enc	Binary	M=1, F=0
Income_enc	Ordinal	Low=0, Medium=1, High=2
AgeGroup_enc	Ordinal	Young=0, Middle=1, Senior=2
IsHighRisk	Binary	Engineered: low login + complaints flag
HasServiceRecord	Binary	Engineered: ever contacted support
MS_Married, MS_Single, MS_Widowed	Binary (one-hot)	MaritalStatus — Divorced is baseline
SU_Online Banking, SU_Website	Binary (one-hot)	ServiceUsage — Mobile App is baseline
FC_Clothing, FC_Electronics, FC_Furniture, FC_Groceries	Binary (one-hot)	FavCategory — Books is baseline
ChurnStatus	Binary (TARGET)	0 = retained, 1 = churned

4.3 Recommended Next Steps

- Address class imbalance (80/20 split) using SMOTE oversampling or class_weight='balanced' in the model — do not evaluate with accuracy alone.
- Train baseline models: Logistic Regression, Random Forest, and XGBoost. Compare on AUC-ROC and F1-score.
- Use feature importance scores from Random Forest or XGBoost to identify and potentially drop low-importance features.
- Consider cross-validation (k=5) to ensure model performance is not sensitive to train/test split.
- The cleaned dataset (Churn_Model_Ready.xlsx) is included as a separate deliverable and is ready for immediate model training.